

Model 1

Mara Dukuly and Adi Suresh

September 30, 2026

1 Motivation

- **Scenario:** Suppose we have a country affected by some conflict.
- **Assumption:** We can provide them with the money to rebuild their country after the war, and they have their pre-war levels.
- **Example:** The number of schools, hospitals, etc.
- **Observation:** War-affected countries seem to have persistent effects in terms of long-term development (i.e. Liberia, Sierra Leone, Sudan, Eritrea, Ethiopia, etc.)
- **Question:** What is missing?
- **Answer:** Human capital and skill formation.
- Over some period of conflict, an individual does not build their “set” of skills and therefore loses potential gains from skill formation they would otherwise have built during that period.
- We assume the individual had, or would have built, some sort of skill-capital during the war period through:
 - Primary school education
 - Parental investment
 - Work experience
- This loss then poses an inter-generational issue.

How can we aggregate individuals’ conflict exposure and skill formation losses to explain current macroeconomic outcomes?

Develop a theoretical and empirical model that captures the following:

1. Capture learning-by-doing losses and skill formation as human capital in a CES production function.
2. Measure heterogeneous effects in agents as aggregated human capital losses across multiple periods.

2 Model

Each time period t is a 3-year period until the individual is 47. Let a represent the age of the agent in the current period, where A is the age in period t_N , the age at death. Then $a \in [0, 47]$ and $t \in [0, 16]$.

Parameters.

- β : discount factor.
- γ : coefficient of relative risk aversion.
- α : capital share.
- δ : depreciation rate of physical capital.
- $\Omega_{a,t}$: productivity of the skill-formation technology.
- $\phi_{a,t}$: weight on investment.
- $\theta_{a,t}$: weight on own human capital.
- $1 - \phi_{a,t} - \theta_{a,t}$: weight on parental human capital.
- ρ_t : substitution parameter; the elasticity of substitution is $1/(1 - \rho_t)$.
- A : age at death (number of ages) in the sums and in the $\bar{H}$ equation.
- T : final period.

The agent solves the following sequential problem:

$$\max_{\{C_{a,t}, X_{a,t}, I_t, K_{t+1}, H_{a,t+1}\}_{t=0}^T} \sum_{t=0}^T \beta^t \sum_{a=0}^A \frac{C_{a,t}^{1-\gamma}}{1-\gamma} \quad (1)$$

subject to, for all $t = 0, \dots, T$ and $a = 0, \dots, A$:

$$\sum_{a=0}^A C_{a,t} + \sum_{a=0}^A X_{a,t} + I_t + K_t \leq Y_t \quad (\text{resource constraint}) \quad (2)$$

$$Y_t = AK_t^\alpha (H_t L_t)^{1-\alpha} \quad (\text{output}) \quad (3)$$

$$K_{t+1} = (1 - \delta)K_t + I_t \quad (\text{capital accumulation}) \quad (4)$$

$$H_{a+1,t+1} = \Omega_{a,t} \left[\phi_{a,t} X_{a,t}^{\rho_t} + \theta_{a,t} H_{a,t}^{\rho_t} + (1 - \phi_{a,t} - \theta_{a,t}) H_{P,t}^{\rho_t} \right]^{1/\rho_t} \quad (\text{skill formation}) \quad (5)$$

$$\bar{H}_{t+1} = \frac{1}{A} H_0 + \frac{A-1}{A} \bar{H}_t, \quad H_t \equiv \bar{H}_t \quad (\text{average human capital}) \quad (6)$$

$$C_{a,t} \geq 0, \quad X_{a,t} \geq 0, \quad K_{t+1} \geq 0, \quad (7)$$

with $\rho_t \in (-\infty, 1)$, given the initial conditions K_0 and $\{H_{a,0}\}_{a=0}^A$. When agents are aged 12–15, their human capital at that point is taken as the entry level $H_0 = H_{a,t}$ in the average human capital equation.

Variables.

- *Choice variables*: consumption $C_{a,t}$; skill investment $X_{a,t}$; physical investment I_t .
- *State variables*: physical capital K_t ; human capital of an agent of age a , $H_{a,t}$; average human capital $\bar{H}_t$.
- *Other endogenous variables*: output Y_t ; parental human capital $H_{P,t}$; entry-level human capital H_0 .
- *Exogenous*: labor L_t ; TFP A in the output equation.